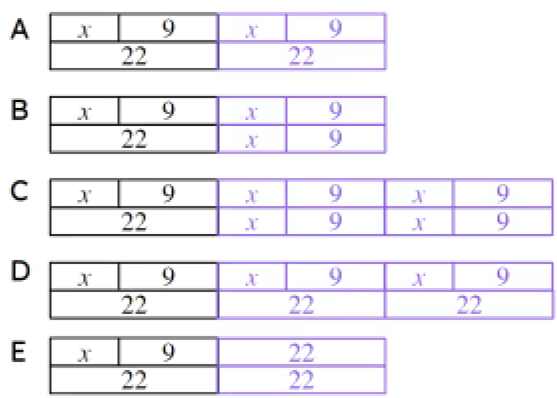
Securing understanding of equality

1 Which of these is the best definition for like terms? Tick **1** correct answer

- [ ] Terms which when added, sum to zero.
- [ ] Terms with the same coefficient for each variable.
- [x] Terms with the same set of variables and corresponding exponents.
- [ ] Identical terms which have the same coefficient and same variables.
- [ ] Terms with the same variable and different exponents.

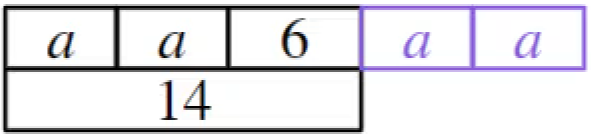
2 These 5 bar models represent operations performed on the equation $x + 9 = 22$. Match the bar models labelled A, B, C, D and E with the equivalent equation they represent. Write the correct letter in each box



a	A
b	B
c	C
d	D
e	E

b	$x + 9 + x + 9 = 22 + x + 9$
c	$x + 9 + 2x + 18 = 22 + 2x + 18$
a	$2(x + 9) = 2(22)$
d	$3(x + 9) = 3(22)$
e	$x + 9 + 22 = 22 + 22$

3 This bar model represents the initial equation $2a + 6 = 14$. What needs to be done to the bottom bar to maintain equality? Tick **1** correct answer



- [ ] $\times 2$
- [ ] $+ 14$
- [x] $+ 2a$
- [ ] $+ 6$
- [ ] $\times a$

4 If $x + 5 = 2$, which of these equations have maintained equality? Tick **3** correct answers

- [ ] $2x + 5 = 4$
- [x] $2x + 5 = x + 2$
- [x] $x + 3 = 0$
- [ ] $x + 12 = 10$
- [x] $\frac{x + 5}{2} = 1$

5 If $a = b$, which of the following are true? Tick **2** correct answers

- [x] $a - c = b - c$
- [ ] $ac = \frac{b}{c}$
- [ ] $a + c = b - c$
- [x] $ac = bc$
- [ ] $a + c = b + d$

6 Which expression could fill the blank on the right hand side of this equation to maintain equality? Tick **3** correct answers

$$a + b = 5$$
$$a + b + a + b + a + b = \_\_\_\_\_\_\_\_$$

- [ ] 10
- [x] 15
- [ ] 5 + a + b
- [x] 5 + a + b + a + b
- [x] 2a + 2b + 5